

Containers and Kubernetes TASK – 4

1.Create a new public GitHub repository named insta-caption-bot and add a simple Dockerfile that builds a Node.js app which prints 'Hello, Instagram!' when run.

Ans :

```
gh repo create insta-caption-bot --public --clone
```

```
cd insta-caption-bot
```

```
cat > package.json <<'EOF'
```

```
{  
  "name": "insta-caption-bot",  
  "version": "1.0.0",  
  "main": "index.js",  
  "scripts": {  
    "start": "node index.js"  
  }  
}
```

```
EOF
```

```
cat > index.js <<'EOF'
```

```
console.log("Hello, Instagram!");
```

```
EOF
```

```
cat > Dockerfile <<'EOF'
```

FROM node:22-alpine

WORKDIR /app

COPY package.json ./

COPY index.js ./

CMD ["npm", "start"]

EOF

docker build -t insta-caption-bot .

docker run --rm insta-caption-bot

git add .

git commit -m "Add Node.js Docker app"

git push -u origin main

2. Write a GitHub Actions workflow file

(.github/workflows/docker-build.yml) that automatically builds the Docker image from your repo whenever you push code to the main branch.

Ans :

mkdir -p .github/workflows

cat > .github/workflows/docker-build.yml <<'EOF'

name: Docker Build

on:

push:

branches:

- main

jobs:

docker-build:

runs-on: ubuntu-latest

steps:

- name: Checkout repository

uses: actions/checkout@v4

- name: Build Docker image

run: docker build -t insta-caption-bot .

EOF

git add .github/workflows/docker-build.yml

git commit -m "Add Docker build GitHub Actions workflow"

git push origin main

**3.Modify your workflow to push the built Docker image to Docker Hub using your Docker Hub credentials as GitHub Secrets.

Hint: Use the**

docker/login-action and docker/build-push-action in your workflow.

Ans :

```
mkdir -p .github/workflows
```

```
cat > .github/workflows/docker-build.yml <<'EOF'
```

```
name: Build and Push Docker Image
```

```
on:
```

```
  push:
```

```
    branches:
```

```
      - main
```

```
jobs:
```

```
  docker:
```

```
    runs-on: ubuntu-latest
```

```
    steps:
```

```
      - name: Checkout repository
```

```
        uses: actions/checkout@v4
```

```
      - name: Log in to Docker Hub
```

```
        uses: docker/login-action@v4
```

```
        with:
```

```
username: ${{ secrets.DOCKERHUB_USERNAME }}
```

```
password: ${{ secrets.DOCKERHUB_TOKEN }}
```

```
- name: Build and push Docker image
```

```
uses: docker/build-push-action@v7
```

```
with:
```

```
context: .
```

```
push: true
```

```
tags: ${{ secrets.DOCKERHUB_USERNAME }}/insta-  
caption-bot:latest
```

```
EOF
```

```
git add .github/workflows/docker-build.yml
```

```
git commit -m "Push Docker image to Docker Hub"
```

```
git push origin main
```

4.Update your Node.js app to print a random Zomato-style food review (pick from 3 hardcoded options) instead of 'Hello, Instagram!', and confirm the new message appears in the Docker image built by your CI/CD pipeline.

Ans :

```
cat > index.js <<'EOF'
```

```
const reviews = [
```

```
"Amazing pizza! Crispy crust, generous toppings, and delicious!",
```

```
"Juicy burger with fresh ingredients — absolutely loved it!",
```

```
"Flavorful noodles, perfectly cooked and served hot!"
```

```
];
```

```
const review = reviews[Math.floor(Math.random() * reviews.length)];
```

```
console.log(review);
```

```
EOF
```

```
git add index.js
```

```
git commit -m "Add random Zomato-style food reviews"
```

```
git push origin main
```

```
echo "Waiting for GitHub Actions to build the new image..."
```

```
sleep 5
```

```
docker pull $(git remote get-url origin | sed -E  
's#https://github.com/([^/]+)/.*#\1#')/insta-caption-bot:latest
```

```
docker run --rm $(git remote get-url origin | sed -E  
's#https://github.com/([^/]+)/.*#\1#')/insta-caption-bot:latest
```

5. Use GitHub Copilot or ChatGPT to help you write or debug your GitHub Actions workflow file, and paste the prompt you used along with the final working YAML.

Ans :

name: Build and Push Docker Image

on:

push:

branches:

- main

jobs:

docker:

runs-on: ubuntu-latest

steps:

- name: Checkout repository

uses: actions/checkout@v4

- name: Log in to Docker Hub

uses: docker/login-action@v3

with:

username: \${{ secrets.DOCKERHUB_USERNAME }}

password: \${{ secrets.DOCKERHUB_TOKEN }}

- name: Build and push Docker image

uses: docker/build-push-action@v6

with:

context: .

push: true

```
tags: ${{ secrets.DOCKERHUB_USERNAME }}/insta-  
caption-bot:latest
```